

Chapter 16.5: Surfaces

1. Consider the surface cut from the parabolic cylinder $y = 4 - x^2$ by the planes $z = 0$, $z = 2$, and $y = 0$. Sketch S and find a parameterization of S .
2. Consider the surface left in the hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$ after cutting off the lower part between the planes $z = 0$ and $z = \sqrt{3}$ from the hemisphere. (In short, S is the surface which is the part of $x^2 + y^2 + z^2 = 4$ above $z = \sqrt{3}$). Sketch S , parametrize S and find the surface area of S .
3. The tangent plane at a point $P_0 = (f(u_0, v_0), g(u_0, v_0), h(u_0, v_0))$ on a parameterized surface $\mathbf{r}(u, v) = \langle f(u, v), g(u, v), h(u, v) \rangle$ is the plane through P_0 normal to the vector $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)$.

Use this to find an equation to the tangent plane of the surface parameterized by $\mathbf{r}(r, \theta) = (r \cos(\theta))\mathbf{i} + (r \sin(\theta))\mathbf{j} + r\mathbf{k}$, $r \geq 0$, $0 \leq \theta \leq 2\pi$ at the point where $(r, \theta) = (2, \pi/4)$.

What is a Cartesian equation for this surface? Sketch it and the tangent plane.

4. Find the area of the part of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 1$.
5. Find the area of the surface cut from the “nose” of the paraboloid $x = 1 - y^2 - z^2$ by the yz -plane.

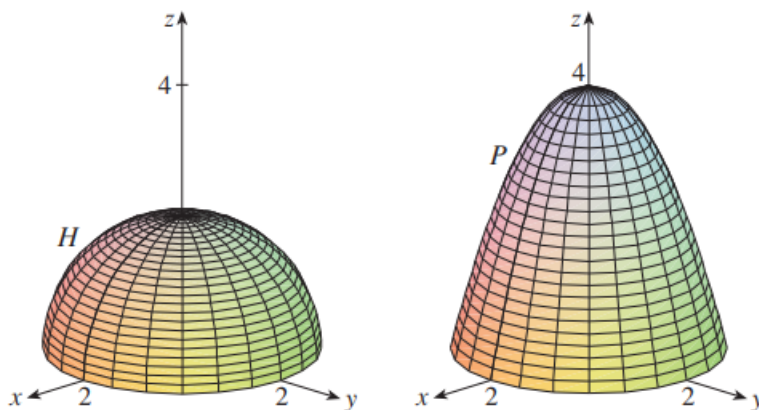
Chapter 16.6: Surface Integrals

1. Integrate $f(x, y, z) = yz$ over the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the cone $z = \sqrt{x^2 + y^2}$.
2. Find the flux of the field $\mathbf{F}(x, y, z) = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$ across the surface S which is the boundary of the solid half-cylinder $0 \leq z \leq \sqrt{1 - y^2}, 0 \leq x \leq 2$, with the outward orientation.
3. A fluid has density 870 kg/m^3 and flows with velocity $\mathbf{v} = \langle z, y^2, x^2 \rangle$, where x, y, z are measured in meters and the components of $\mathbf{v}$ in meters per second. Find the rate of flow outward through the cylinder $x^2 + y^2 = 4, 0 \leq z \leq 1$.

Chapter 16.7: Stokes' Theorem

- Let H be the hemisphere and P be the portion of a paraboloid shown below. Use Stokes' Theorem to explain why, if $\mathbf{F}$ is a vector field on $\mathbb{R}^3$ whose components have continuous partial derivatives, we must have

$$\iint_H (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma = \iint_P (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma.$$



- Use Stokes' Theorem to evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma$, where $\mathbf{F} = \langle x^2 z^2, y^2 z^2, xyz \rangle$ and S is the part of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 4$, oriented upward.
- A particle moves along line segments from the origin to the points $(1, 0, 0)$, $(1, 2, 1)$, $(0, 2, 1)$, and back to the origin under the influence of the force field

$$\mathbf{F}(x, y, z) = z^2 \mathbf{i} + 2xy \mathbf{j} + 2y^2 \mathbf{k}.$$

Find the work done by the field on the particle.